Lab: Bayesian Mechanics and the Free Energy Principle

> **Learning Goal:** Analyze the mathematical foundations of Bayesian mechanics, working through the particular partition, Helmholtz decomposition, and path integral formulations.

Part 1: Particular Partition Analysis

Exercise: For each of the following systems, identify the four partition states:

System	External (η)	Sensory (s)	Active (a)	Internal (μ)
Cell	Nutrient concentrations, toxins	Membrane receptors, ion channels	Metabolic outputs, motility	Intracellular signaling, gene expression
Brain (neuroscience)	World state	Sensory neurons (retina, cochlea)	Motor neurons, glands	Cortical neurons, subcortical circuits
Thermostat	Room temperature	Temperature sensor	Heater on/off switch	Electronic comparator circuits
Bacterium	Chemical gradient	Chemoreceptors	Flagellar motors	Intracellular signaling cascades

For the cell example, verify: (a) External states influence sensory but not internal states directly. (b) Internal states influence active but not external states directly. (c) The Markov blanket is formed by sensory + active states.

Write your response here

Part 2: Helmholtz Decomposition Exercise

> **Learning Goal:** Decompose dynamics into dissipative and solenoidal components.

Exercise: Consider a 2D system with states $x = (x_1, x_2)$ and dynamics:

$dx_1/dt = -x_1 + 2x_2 +$ noise $dx_2/dt = -2x_1 - x_2 +$ noise

1. Identify the dissipative component (symmetric part of the drift matrix): $\Gamma = ?$
2. Identify the solenoidal component (antisymmetric part): $Q = ?$
3. What does the dissipative component do? (Gradient descent toward the mode)
4. What does the solenoidal component do? (Rotational flow around the mode)
5. Sketch the combined flow field. Note: the system spirals toward the fixed point.

Relate this to neural dynamics: the dissipative component = inference (converging to a belief), the solenoidal component = exploration (oscillatory dynamics).

Write your response here

Part 3: NESS and Inference

> **Learning Goal:** Understand when physical dynamics are equivalent to inference.

Exercise: Write out the conditions for the Free Energy Lemma to apply:

1. **Stationarity**: The system must be at NESS — time-invariant probability density

2. **Markov blanket**: The particular partition must hold
3. **Smooth dynamics**: The drift must be differentiable
4. **Ergodicity**: The system must visit its typical states

Now consider a violation of each condition: What happens to the inference interpretation if...

- The system is not at steady state (e.g., developing embryo)?
- The Markov blanket leaks (internal states directly influence external states)?
- The dynamics are discontinuous (phase transitions)?
- The system is not ergodic (irreversible transitions)?

Write your response here

Part 4: Path Integral Analysis

> **Learning Goal:** Work with path-integral formulations of free energy.

Exercise: Consider a system evolving from state x_0 at t=0 to state x_T at t=T:

1. Write the action functional $S[x(t)] = \int_0^T L(x, dx/dt)\, dt$
2. The most likely path minimizes S. How does this connect to the principle of least action?
3. Why must we consider *distributions over paths* rather than single paths?
4. How does path-integral free energy F[q(x(t))] generalize instantaneous free energy?
5. What is the "temporal cost" of changing beliefs quickly vs. slowly?

Write your response here

Part 5: Critique and Limits

Learning Goal: Critically evaluate the scope and limits of Bayesian mechanics.

Exercise: Evaluate these critiques:

Critique	Strength (1-5)	Your Response
"FEP is unfalsifiable — any system at NESS can be described this way"		Is mathematical equivalence a scientific claim?
"Markov blankets in biological systems are approximate, not exact"		How does this affect the formal results?
"The dual-aspect interpretation overrelates inference and physics"		Where does the analogy break down?
"Bayesian mechanics is just a re-description, not an explanation"		What explanatory work does it do?
"The gauge freedom makes the beliefs unidentifiable"		Does gauge freedom undermine the interpretation?

Write a 300-word analysis: What is the epistemological status of the FEP? Is it a law, a principle, a framework, or a tautology?

Write your response here

Lab Summary

Part	Skill Practiced	Key Concept
1	System analysis	Particular partition identification
2	Mathematical analysis	Helmholtz decomposition
3	Formal conditions	NESS and inference equivalence
4	Advanced mathematics	Path integral formulation
5	Critical evaluation	Epistemological status of FEP